

Your Migration Map – SQL Server DW to a BigQuery Medallion Lakehouse

Your end-to-end target architecture on one page, built up piece by piece across the two days and completed at the Day-2 capstone. This is the card to print and pin by your desk.

The whole flow

SOURCES INGEST — land it once, in one place

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SQL Server databases ---> Datastream: log-based CDC – backfill + continuous changes.
(case mgmt, finance, ...) No triggers, no polling, no nightly window.
SQL Server is a supported source.

County file drops ---> Files land in a GCS bucket (Storage Transfer or scheduled copy), (SFTP shares, counties) then LOAD DATA with a PINNED schema – never autodetect for anything recurring.

SaaS applications ---> BigQuery Data Transfer Service: managed, scheduled pulls.

Ad-hoc / one-off files ---> Console upload, bq load, or LOAD DATA.

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BRONZE — raw, immutable, replayable

GCS buckets: Parquet in Hive layout (county=X/period=YYYY-MM/), kept as-landed forever.

BigLake / Iceberg tables over the files + Datastream-landed raw tables beside them.

Queryable from minute one; the source of every future replay.

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Dataform staging + incremental models, with assertions
(all .sqlx in git, reviewed in PRs — your CI/CD discipline)

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SILVER – cleaned & conformed

Deduped, typed, joined to reference data; one staging model per source variant.

Assertions catch schema drift and surprise NULL rates before your analysts do.

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Dataform release configuration (Silver -> Gold, on a schedule)

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GOLD – data marts, the single source of truth

BigQuery mart tables, partitioned by date, clustered by the columns you filter on.

Governed: IAM, Knowledge Catalog, sensitive-data protection for PII.

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Google BigQuery connector (Get Data in Power BI)

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SERVING — the Power BI you already have

Same reports, authors, workspaces, DAX. Import + scheduled refresh by default;

DirectQuery only when a mart genuinely needs intra-day freshness.

ORCHESTRATION SPINE (runs the whole flow, everything defined in git):

Cloud Scheduler (= SQL Agent) -> Dataform release runs -> Power BI refresh follows the release.

Datastream runs itself — M&O watches a freshness metric instead of restarting a 2 AM job.

Stage by stage

Stage	GCP tool(s)	What it does	Replaces in your current stack
Ingest — databases	Datastream	Log-based change data capture: initial backfill plus continuous inserts/updates/deletes into BigQuery	Nightly extract jobs, replication
Ingest — files	GCS + Storage Transfer , then LOAD DATA	Drops land in a bucket once; loaded with a pinned schema JSON	SSIS file-sweep packages
Ingest — SaaS	BigQuery Data Transfer Service	Managed, scheduled pulls from SaaS sources	Vendor exports + manual loads
Bronze	GCS (Parquet, Hive layout) + BigLake/Iceberg	Raw landing: immutable, replayable, queryable in place	Staging shares + staging tables
Silver	Dataform staging + incremental models, assertions	Cleaning, conforming, dedupe — with tests in the repo	Transform sprocs / SSIS data flows
Gold	BigQuery marts (partitioned + clustered)	Curated marts — the single source of truth for reporting	Reporting database / mart tables
Serving	Power BI + Google BigQuery connector	Import (default) or DirectQuery against Gold	The reports you already run
Orchestration	Cloud	Schedule →	SQL Agent jobs +

Stage	GCP tool(s)	What it does	Replaces in your current stack
Governance	Scheduler ♦	release →	your CI/CD
	Dataform	refresh, boring on purpose	pipelines
	release configs, in git		
	IAM,	One governed	Your security &
	Knowledge	source of truth;	PII controls,
	Catalog,	PII discovered	centralized
	Sensitive Data	and protected	
	Protection		

Rules of the map

- **Every box gets one tool, not three.** A cell that says "Datastream or DTS or Dataflow" is a deferred decision — acceptable in backlog draft zero, but name it as one.
- **Bronze answers 'what did the source say'; Silver answers 'what does it mean.'** No cleaning rules in Bronze — that separation is what makes reprocessing possible, the thing your current extract jobs can't do.
- **Dataform owns Silver→Gold, and it's in git.** Transforms living anywhere else (heavy scheduled queries, Power Query in Power BI) need a stated reason.
- **Orchestration is boring.** Scheduler + Dataform release configurations cover nearly everything you mapped in class. Composer (managed Airflow) enters when you have many interdependent pipelines or non-BigQuery tasks — not on day one.
- **Power BI is Import-first.** DirectQuery appears only with a stated freshness need, and refresh cadence names the pipeline event it follows ("after the Dataform release"), not a wall-clock guess.
- **Fill in 'first thing that breaks' honestly.** Schema drift, PII, a county that re-sends files, 10× volume — that row becomes your migration risk register.

Dig deeper

- [Datastream](#) — [sources](#) (incl. [SQL Server](#)) · [Storage Transfer Service](#) · [LOAD DATA](#)
- [Iceberg tables](#) · [BigLake](#) · [Partitioned tables](#) · [Clustered tables](#)
- [Dataform](#) · [Scheduled queries](#) · [Cloud Scheduler](#)
- The Dataform repo walked through in class on Day 1: <https://github.com/jwdavis/dataform-demo>
- The last mile in detail: [Power BI ↔ BigQuery connector guide](#)

Part of the Iowa DOM DoIT Data Engineering student pack — see ../README.md.